CMP6200

Individual Undergraduate Project (FYP)

Final Report

New Ways to Play: A Study in Alternative Game Controllers

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Abstract

The project investigates alternative game controllers and their impact on player experience, immersion and accessibility. While traditional controllers such as the PlayStation and Xbox gamepads remain the dominant design for most controllers, emerging technologies offer new ways to play, particularly through motion-based input. A custom motion-based controller was developed using a Raspberry Pi Pico and integrated with a Unity-based game environment. Which was designed to understand how alternative controllers can influence gameplay, like input accuracy, responsiveness and user immersion.

User testing was conducted which collected detailed data through an online form and play testing focusing on the design of the controller, usability, accessibility and how it influenced immersion and experience. The data collected from the users will be anonymised and collected data like controller design, accessibility, usability, responsiveness and immersion.

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1.0 Introduction and Context

This report aims to examine alternative game controllers and their impact on gameplay, immersion and overall player experience. The choice of input device plays a significant role in determining the players' interaction with a game. (Akyaman & Cem Alppay, 2024).

The project will consist of designing and developing a game alongside a custom controller, both which will continually change and improve throughout the development process. This will help explore how hardware design and the game or software can shape the experience with one another. In addition, this report investigates the role of alternative controllers, primarily motion-based controllers. Particularly how they can affect accessibility as well as their potential to broaden participation for players who may not want to or are unable to use a traditional controller.

Furthermore, the report evaluates the hardware advantages and limitations, such as how this may affect performance and sustainability. It also analyses how factors such as layout, ergonomics, and functionality influence player immersion and overall usability. The report will also go into detail about the time and development process of the motion-based input device that was being developed for the project.

Finally, the report will also go into detail about the practical development process, including design ideas, planning and time it took. This includes an overview of the tools, materials and technologies used, as well as reflection on the project timeline

and the challenges encountered during implementation and providing insight into the integration of hardware and software components and the testing results of the final product.

1.1 Problem Statement

Gaming controllers have always had the expectation to create the standard layout and input methods for most people. But this comes with its own issue where it may not cater to everyone, and many players want to explore alternative ways to interact with games to increase their immersion, experience and personal preference.

Furthermore, many may be unable to effectively use the traditional controller due to physical limitations. Research indicates that most traditional gaming controllers are not designed with accessibility in mind, making them difficult to use (Malone, et al., 2023) and potentially losing out on a wider range of players.

This highlights the gap in the market which creates a barrier to entry for small studios and developers to create alternative controllers and address the solution for accessibility and immersive gameplay by being able to feed into the needs of the consumers that can support a wider range of players while still maintaining effective gameplay interaction. This project aims to study and address the issue by developing a custom controller that will be designed along with a game environment, enabling a more interactive and integrated player experience.

2.0 Review of Existing Knowledge

Technical Development

Alternative game controllers have been around for many years from the popularity of motion controls such as the Nintendo Wii and Kinect to more advanced technology like Virtual Reality which allows the user to essentially be in a world of their own which can track full human interaction. A lot of these alternative game controllers still use the traditional buttons and joysticks, but more have started using as little as possible and mostly dependent on motion control and non-traditional inputs.

Immersion

Immersion refers to a person having full engagement on the task they are focused on. This is related to the project because players can be so immersed on the task while interacting with motion-based controller within the game, they can lose awareness perception of the real world while focusing on the task environment (Jennett, et al., 2008).

There are different forms of immersion such as sensory, challenge based, narrative immersion etc. But the form of immersion that will be focused on in this report will be how alternative game controllers can help with the players' experience and immersion within video games. The effectiveness of motion-based

controllers depends on how natural the inputs feel when playing a game. When the controls feel like an action a person would do in the real world, research shows that players can interact more effectively and naturally, which enhances immersion (Skalski, et al., 2010). For the motion-based controller the player would be in a shooting arcade stance with the arms so the testers would hold the controller straight forward in an aiming position. However, if the controls aren't well implemented or does not align with the game mechanics, it can lead to frustration and reduced overall immersion and enjoyment.

Alternative Game Controllers

Immersion is a focal point in the game industry whether that's how often players can turn their devices on to play a game or how long they can sit and play. Games that offer high level of immersion have taken the consideration to the design of their input systems. Research by (Thorpe, et al., 2011) found that alternative game controllers can increase the player immersion and experience when the controller is suited to the game mechanics. Their findings from their project Zion showed that there was already familiarity and controller suitability had stronger influence on immersion than if they used motion-based input. While the Nintendo Wii introduced a new level of technology with the introduction of motion control it was not automatically perceived as more immersive than the traditional controllers such as the Xbox 360 gamepad (Thorpe, et al., 2011). These findings show that the effective controller design depends on how naturally the input methods map to the actual game that is being played. Motion-based controller similar the artefact that has been created for the project was mostly well received because of the game environment it was made around; it supported the idea of motion movement as well as arcade style shooting.

These are aspects that a lot of game studios have in mind when developing their games. One of the most important aspects is the input system that allows players to be immersed in the games they play. Having a comfortable controller is just as important as being able to play the well-made game.

Accessibility

Many people that enjoy games have disadvantages and look for alternative controllers that will help them enjoy the game they want to play. This is why accessibility is such an important part of gaming and is required to be considered allowing everyone to enjoy games the same way, as not all players are able to use traditional controllers such as the standard joystick and button layouts as seen on most gamepads. Players with the physical limitations may require alternative game controllers to play games effectively so the use of motion-based controls can be very beneficial and still immerse players with disadvantages in any game a drastically improve experience. Research shows that individuals with limited mobility often rely on multi-modal input methods rather than a single traditional device (Wentzel, et al., 2022).

This shows the importance of alternative game controllers, like the motion-based system that has been created part of the artefact which can provide more ways to play. This opens to a wider range of players to engage with and enjoy the experience. In addition, alternative controllers can further enhance and improve overall player experience because some may want to explore different ways to game rather than using traditional inputs.

2.1 Critical Analysis and Gap Identification

This report so far has covered alternative game controllers and how and why they are used to potentially improve player experience and immersion. However, many of these studies emphasise on positive outcomes but overlook drawbacks. For instance, (Thorpe, et al., 2011) found that alternative input methods can improve immersion only when they align with the game mechanics, yet their findings are based on a small sample and a simple prototype game, limiting broadness.

Another key gap is the limited exploration of negative user experiences. While some studies acknowledge that alternative controllers can feel gimmicky when poorly matched to the game (Thorpe, et al., 2011), they do not fully examine player frustration or its impact on constant use. Additionally, research often assumes able-bodied users, with little attention to accessibility or inclusive design.

Further key gaps that aren't discussed in the cites are;

- The lack of consumer needs most people prefer to use regular gamepads instead of alternative game controllers like motion controls when gaming.
- Environmental impacts and how creating these alternative game controllers may cost a lot to produce and contain a lot of unnecessary materials.
- Hardware Complexity a lot of users may not be able to understand the technology and still want to use them

The project artefact created a much simpler motion-based controller where the testers were able to comfortably use the controller without issue, the motion system of the artefact also allowed to interaction outside the game which meant they weren't limited to only the game environment. Making the player experience and immersion much more enjoyable and usable.

2.2 Project Justification

It is important to build on the gaps that are in the research such as lack of studies on environmental impacts, potentially wasting materials to produce these high-end hardware and cost of production. Current studies highlight the potential of motion and VR controls to enhance immersion but also suggest that their effectiveness depends on how well they are implemented and aligned with the gameplay (Thorpe, et al., 2011). Poorly designed control mappings or unfamiliar designs can reduce usability and negatively impact the player experience. Furthermore, accessibility plays an important role in being able to identify and take action to produce a product where anyone can use an alternative controller to experience games, is just as important.

When designing the motion-based controller and the game environment, production of cost and the material used was taken into consideration. Research into consumer electronics highlighted that environments impact of hardware production, including the material use and electronic waste, is often overlooked in the design of gaming peripherals. (Akyaman & Cem Alppay, 2024). The core idea of this project was to fill the gaps that a lot of the research have no considered or not investigated.

This helped ensure that the controller works effectively and helped players understand the hardware behind these features such as motion controls that can support players who may not be able to or want to use the traditional joystick or button on gamepads.

This was achieved by using a Raspberry Pi Pico W and an MPU6050 which helped produce a motion-based controller to showcase how motion-based controls work as well as the advantages and disadvantages.

The outcome of the project is to showcase how alternative game controllers can help players become more familiar and learn to better understand non-traditional input method, such as motion-based interaction. Research by (Skalski, et al., 2010) demonstrates that controllers with more natural mappings improve player interaction with the game enjoyment. This indicates that that exposure to alternative controllers can help them prepare for emerging technologies such as virtual reality and motion-controlled systems. Additionally, the expansion of alternative input methods like the artefact that has been created during the project can highlight opportunities for further development in game hardware and interaction design, encouraging continued research into improving accessibility and usability for player experience.

3.0 Project Aims, Objectives, and Scope

During this project there will be aims, objectives and scopes that will need to be followed to ensure the project goes all as planned. This will be done with bullet points in each section showing what will be done to ensure the best possible result for the overall project success.

3.1 Project Aim

The aim of this project is to develop an alternative game controller which will be motion-based, as well as a game within unity that will support the controller to help capture the mechanics and how alternative controllers can work and be beneficial.

3.2 Project Objectives

- **Objective 1:** Research focused to establish knowledge foundation
- **Objective 2:** Technical implementation building the input system
- **Objective 3:** Gameplay design integration, creating the game around the input system.
- **Objective 4:** Prototype completion, have a playable build
- **Objective 5:** User testing and evaluate experience
- **Objective 6:** Analysis and documentation

3.3 Project Scope

This project will concentrate on alternative input methods which will help the user capture the full experience of the hardware, such as motion control. Another focus will be researching and prototyping while also developing a game that will help understand this hardware. This project will not be developing multiple input systems or full-scale commercial games. It is purely for analysing how alternative game controllers can shape player experience and immersion and improvements that can be made.

4.0 Project Design

The steps that will be taken are very simple; the product will be a fully working controller along with a game to study alternative game controllers. This will be done by using the current knowledge I have of this study as well as the skills I have learnt to design and develop to bring the best possible results. During the project there will be phases that will keep everything organised and on track. This will be done by using tables and notes like Gantt Charts, GitHub version

control, Kanban and appropriate planning. The tools used will be purchased online as well as the parts required to build the controller but to build the controller heavy planning went through ahead of time so that the project didn't go over budget as well as the best possible results. In addition, software tools that are free such as Unity and Thonny will be used to develop a game that can simultaneously work with the controller to showcase its capabilities.

4.1 Design and Methods

The overall approach to this project will be taken very cautiously and will involve multiple experiments and development phases which will help produce the best results. Designing and prototyping an input system game controller along with a software integration will clearly demonstrate the results of the alternative game controller.

Development of the Unity game will be used as a test environment.
The aim is to evaluate player immersion and experience using the custom designed controller this will be done through multiple in person tests where the players will use the controller as well as take part in an online survey which will help gather results. Even though the project requires a functional build of the input system and unity game, the results still require statistical analysis and data gathering to show the project works and provides the required goal to achieve immersion and player experience.

This project will involve gathering data such as quantitative data like response times, reliability, number of successful inputs. Data qualitative data will also be used, such as user feedback, observation notes about how the player felt and the ease of use and immersion. Design data will be considered, determining if the input system writing schematics are effective and ensuring the sensor calibrations and ensure Unity input mappings are correctly working as intended.

When designing and building the custom game controller, the Raspberry Pi Pico W for prototyping will be used, as it effectively gives all the parts required to design this project, offering a variety of wires, sensory technology, circuit design on breadboard and code in Thonny.

Along with the game controller, the Unity-Based game will be needed to work simultaneously along with the custom based controller using Visual studio C# and Thonny to create gameplay responding to the inputs. Using custom input mapping with Unity built in Input Map should help with connecting the hardware signals to in game actions.

4.2 Justification of Methods

The approach that has been decided during the development of this project has been picked out carefully to ensure the best possible results. Having players be able to test the controller in person with the game that has been made will be a great way to have accurate and precise feedback which will be evaluated, furthermore having play testers take part in an online survey right after will ensure a highly accurate data collection and responses. Having physical one to one testing's where the project can be seen completed by a person without any altering answers or fake feedback is a much better way to test especially for the controller, if the tests were done online the testers would not be able to use the controller as well as non-accurate data because testers may alter answers online and unreliable people.

In addition, using the Raspberry Pi Pico is a reliable and cheap product along with the MPU6050 to ensure a cheap and simple motion-based controller to showcase the capabilities of alternative game controllers. As well as choosing to develop the game using Unity and C# because of its ease of access and knowledge required to develop.

5.0 Artefact Design

The artefact design was inspired by the WII remote it is essentially a handheld device that is placed in a rectangular 3D printed case with the top being a separate part that sits on top to cover the components within the case. The case has rounded corners with smooth surface to ensure the user is comfortable when holding the controller.

For the controller I decided to use green tapes to connect the case and the lid together to give ease of access to the inside components for replacements and upgrades as well as give a simple design such as improved grip so the device can be stable when moving around and holding. As you would hold any other remote the thumb naturally rests on top of the case where the button is just a little bit above the middle of the case.

The case was designed using a software called FreeCad and the material used was a green PLA filament through the Bambulab P1s 3D printer using a 0.4 mm nozzle. The inside of the case contains the electronic components such as the breadboard, Pi Pico W H and the MPU6050. The overall shape of the case was designed with the intention of it looking like a remote and support the use of one-handed motion control as well as it being portable to carry around.

Figure 2. Design of case with Lid

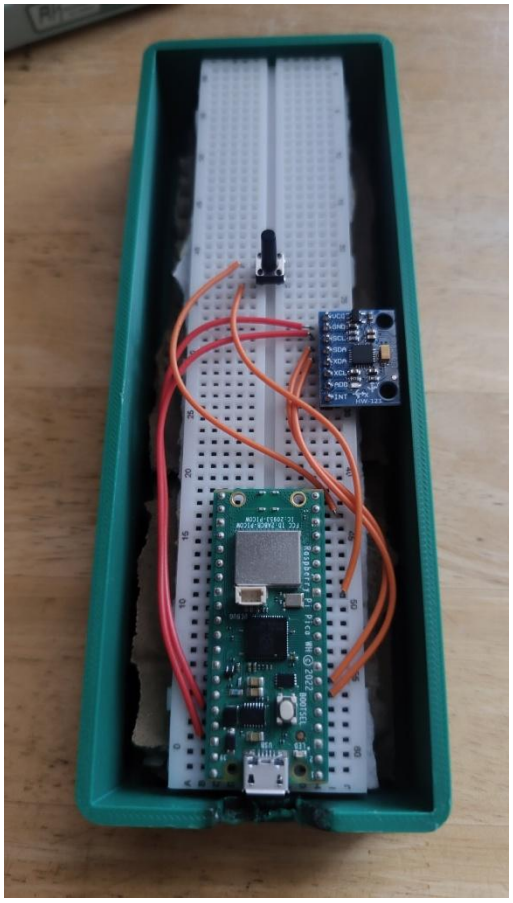

Figure 1. Components within the Case

5.1 Design Specifications

Specification	Requirements / Constraint
Portability	The artefact must be compact and lightweight enough to be comfortably used with one hand
Ergonomics	The design must allow a natural thumb position and secure grip when holding the controller.
Durability	The case must protect the internal electronics from impacts and handling.
Material	The casing must be manufacturable using 3D printing with PLA filament.
Size Constraints	The artefact dimensions must fit within the available 3D printer build size.
Internal Space	The enclosure must provide sufficient room for the internal electronics and wiring

Accessibility	The design must allow access to charging ports or internal components for maintenance
Safety	Sharp edges and exposed wiring must be avoided during normal use
Usability	The device must support simple one-handed interaction
Manufacturing	The design should minimise print complexity and material usage.

Table 1 - Table of Design Specifications

5.1 Artefact's Structure/ Architecture / Framework

The artefact is a handheld motion-controlled device designed to work as a HID (Human interface device) for controlling the computer mouse cursor. It consists of a 3D printed Case for the controller with hardware such as the Pi Pico WH, MPU6050 and jumper wires to connect to each other as well as a 6x6 button to function all together. The case provides support for the components to keep it well protected as well as a comfortable shell for the player to hold.

The Pi Pico uses Thonny as the python programming language and tool to ensure it can read and connect with the MPU6050 giving the ability to motion control the computer cursor.

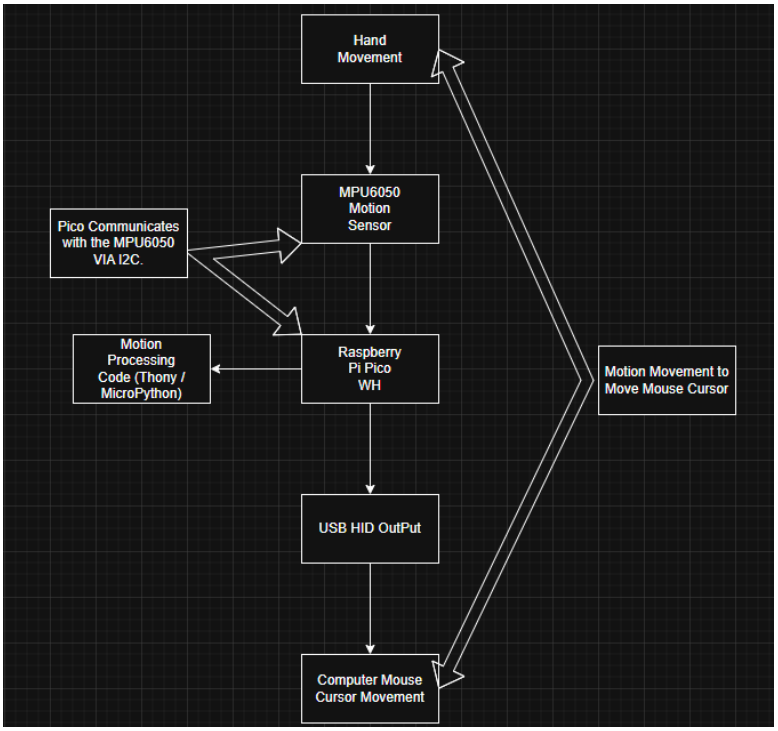

Figure 3. System Architecture

Components	Function
3D Printed Case	Places the components inside and protects internal components while still allowing ergonomic handling.
Raspberry Pi Pico WH	Main Controller for processing sensor data and generating HID Mouse output.
MPU6050	Registers acceleration and rotational movement of the device.
Breadboard	Provides for circuit connections
Jumper Wires	Connect electronic components together
6x6 Push Button	Used for clicking and interaction
USB Connection	Supplies power and transfer HID data to the computer.
Soldering Iron Pen	To connect the pins to the MPU6050
Soldering Iron Metal	Filler metal to join parts together when melting with the Soldering Pen

Table 2. Key Components

Items	Technology / Material
Controller Case	PLA filament manufactured using Bambulab P1S 3D printer
Programming Language	MicroPython (Thonny) and C# (Visual Studio)
Motion Sensor	MPU6050 Accelerometer and gyroscope
Communication Method	USB HID protocol
Embedded System	Raspberry Pi Pico WH Microcontroller

Table 3. Technologies and Materials Used

The MPU6050 constantly measures movement and orientation when the player moves the controller, this data is transferred to the Raspberry Pi Pico WH through wired connections on the breadboard. Connecting the VCC, GND, SCL and SDA from the MPU6050 to the Pico Board which are 3V3 (pin36), GND (pin38), GP4 (pin6) and GP5 (pin7).

The MicroPython Program that is installed and running on the Pico WH ensures it reads the sensor readings from the MPU6050 and converts them into cursor movement. This data is transmitted to the computer through a Micro-USB-B essentially allowing the device to control the cursor movement using motion control. Furthermore, the 6x6 push buttons which were connected using the pins 13 and 19 allows for the player to click the button to interact like a mouse click.

This case has been designed to fit the device inside it and the interaction with button and USB to feel seamless.

6.0 Implementation

The implementation process had multiple phases such as planning, discussing with the supervisor on a weekly basis, research as early as possible so that the correct components were purchased to assemble, program and test the motion-based controller with the game being designed.

The project started with deep analysis, research and planning about alternative game controller, knowing which steps to take to design the artefact and how to present it. Using a Gantt Chart as well as the kanban on GitHub it allowed for clear tracking and planning, starting with researching the type of module kits that allowed for motion tracking because that's what the project was decided on to create a controller that allowed for motion tracking, easy to use and accessible to anyone. Furthermore, also looking into the game engine and programming language to use to design the game environment to test the controller which ended up being Thonny as the controller programming tool and Unity/C# to develop the game.

The motion-based controller went with multiple tests to ensure that it worked with the game, having it first running using an internet connection but later realising to make it more portable the controller had to be further improved to mimic a mouse which will allow it to create a custom device. This was done by switching the Raspberry Pi Pico to use CircuitPython which allows for easy HID support.

Several stages of testing and validation were completed throughout the development. The first stage of testing was ensuring the Raspberry Pi Pico WH and the MPU6050 worked together. Sensor readings were monitored using Debugs in Thonny to confirm accurate motion. Further testing ensured that the responsiveness of the controller was improved to give the player accurate movement and less drift. The 3D case was also designed with the intention of protecting and fitting the controller components inside and giving the player comfortable device to hold.

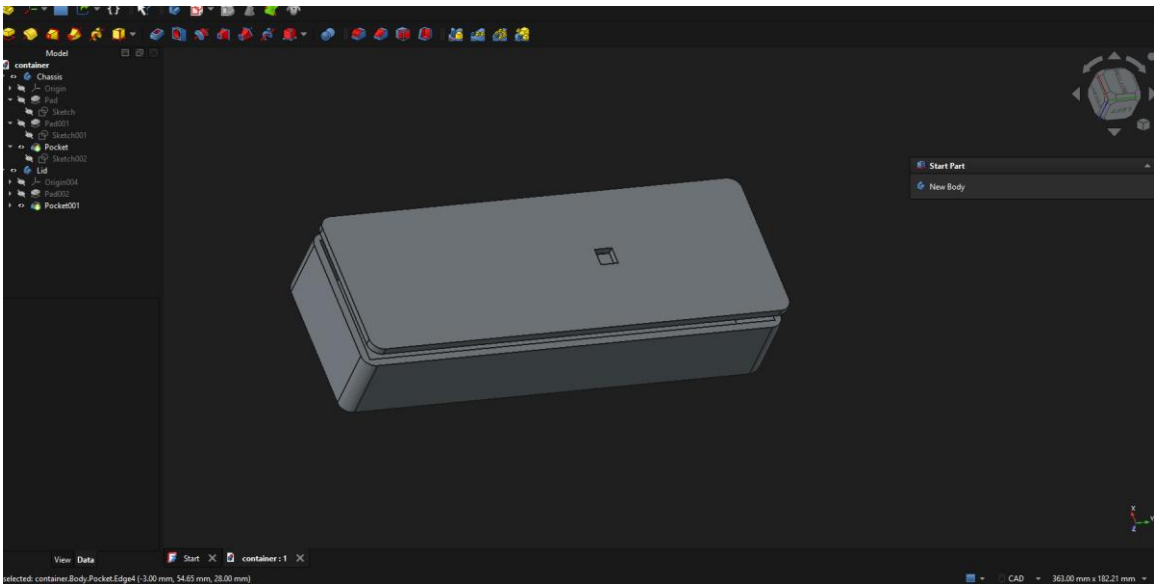

Figure 4. FreeCad Case Design

Final validation and testing involved the artefact fulfilling the project aims and objectives of enabling a motion-based controller that could control the cursor through handheld movement. As well as the controller working in custom based Unity game that allowed the player to play a game using the device.

6.1 Construction of the Artefact

Controller Ideas and analysis

The construction of the artefact involved multiple stages of purchasing and putting together components for both hardware and software. For the controller the first idea was to create a glove with the components on the gloves which meant having to move your whole arm or wrist for motion but after further discussion and analysis that could end up being very ineffective and could lead to arm tiredness and other problems like components not having solid protection. So, the design was changed to be more of a regular controller like the Wii remote or any other TV remote so the player could hold it in their hand and comfortably move it around however they want and just place it down without having to put anything on.

Tools and Languages

The tools that were used during the project were Unity because of its simple and easy to use for game development as well as C# because of prior knowledge and recommendations when building a game especially for an alternative game

controller. For the controller itself Thonny python programming was used to ensure the Pi Pico WH and the MPU6050 communicated together and worked as a HID system.

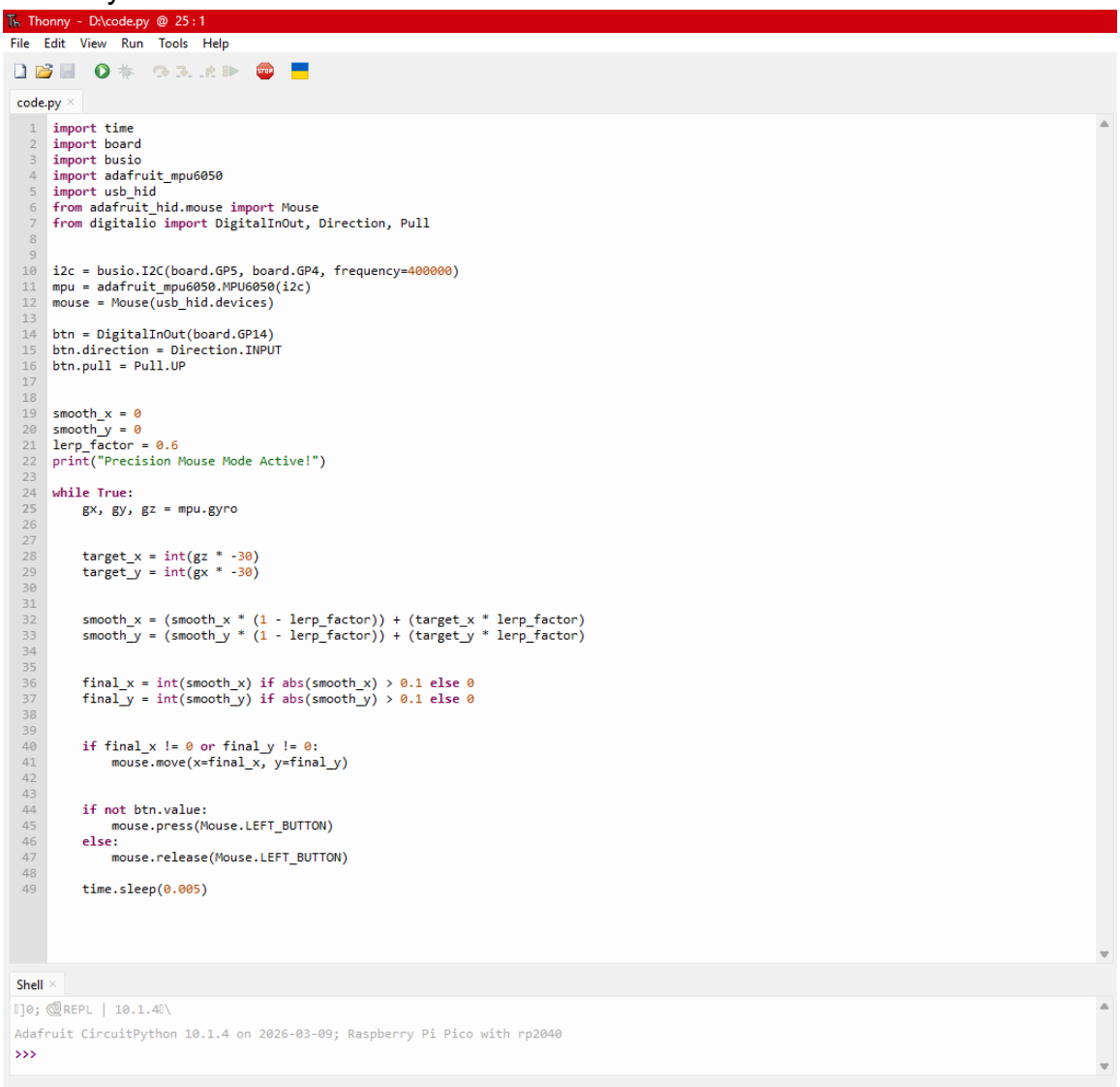

Figure 5. Thonny Pi Pico and MPU6050 Code

The MPU6050 uses a gyroscope sensor that is essentially connected via I2C to detect any movement of the device. Essentially the gyroscope readings are from the gx, gy, gz = mpu.gyro, so to correctly calculate the movement of motion using target_x = int(gz * -30) and target_y = int(gx * -30) the processed movement data is transmitted to the computer using the Micro-USB-B cable to emulate the cursor. Which is essentially how we get the correct motion logic when the player moves the wrist up/down the cursor will on screen goes up/down and same applies with left and right.

In addition, to improve the cursor stability and reduce the sensitive reads from the gyroscope a lerp was implemented. This would help give a more smooth and stable cursor movement and more accurate. A dead zone threshold was also implemented to stop any small movements by gyroscope readings so when the player wasn't moving the controller there would be no unnecessary movements.

A 6x6 push button was also connected to the pins 13 and 19 to allow power and connect. A simple logic that allows for the player to interact like how a left click on a cursor would work.

For the unity game using C# for the scripts a simple arcade shooter game was developed to showcase how motion controls can be used and having a shooting mechanic where aim is important can showcase the controls of the motion-based controller and stress test as well as see how accurate it can be. This was done by having the player shoot at 3 enemies and defeat them before the enemies take the players health down to 0 this was the simple game logic and a good environment for the motion-based controller.

Figure 6. Scene of Match Room

Hardware assembly

After purchasing all the required items such as the breadboard, Flexible jumper wires, 6x6 buttons, Pi Pico WH and MPU6050 the production of the controller began. Starting with the breadboard placing the Pico and ensuring it is securely placed on top, once that was completed soldering the pins to the MPU6050 and

placing it on the breadboard to ensure stability for powering the device as well as keeping it secure on the controller. Lastly placing the buttons on the breadboard as well, once everything was set, all the flexible jumper wires were connected from the Pi Pico to the buttons and MPU6050.

Choosing the MPU6050 because of its cheap price as well as ease of use containing the ability for an accelerometer and gyroscope within it allowing for good readings of movements. The raspberry Pi Pico WH was chosen for its low cost and programmable USB HID technology, small size and MicroPython compatibility.

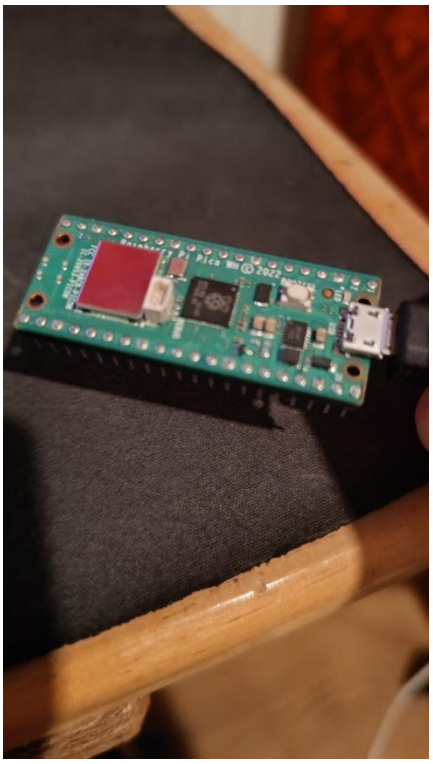

Figure 7. Pi Pico WH

Figure 8. Soldering Iron Pen

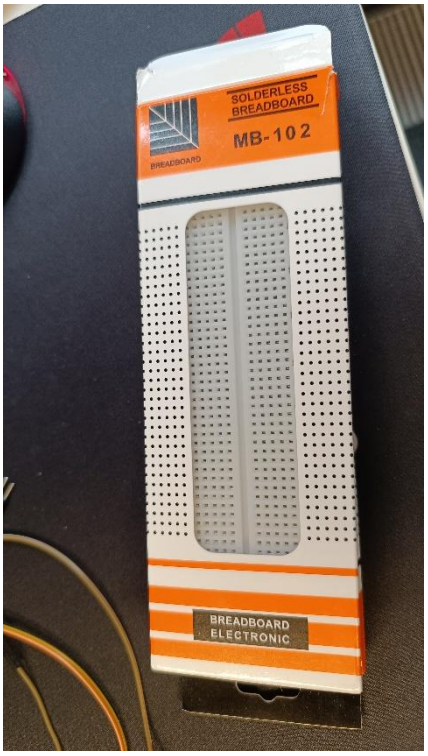

Figure 9. BreadBoard

Figure 10. MPU6050 / Pins

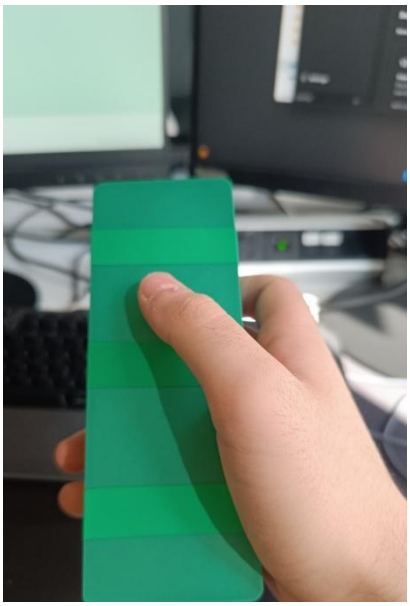

Figure 11. 3D Printed Case and Lid

Figure 12. Jumper Wires / Buttons

Testing and Refinement

Multiple tests were done to ensure that the controller was well connected physically and software, ensuring the cables were well connected and didn't come loose when the player would use motion control. Furthermore, testing was conducted so ensure the controller worked as fast as possible reading data accurately.

Challenges Encountered

Some of the problems that was encountered was the lack of removing delay within the controller so when the player moved the controller to far the cursor would be the same on screen but the controller would be in a completely different direction so the gyro was off centre, this is due the limitations of the MPU6050 being an older hardware high latency and drift. Some other challenges encountered during the project was finding the components that would be required to work as a HID motion-based controller, it took a while to research and make sure that there were no costs wasted and the correct components were purchased. Finding testers was challenging as well because of the time it took to develop the artefact there was little time to gather participants that were willing to take part in the tests and consent to being in the test so the data gathered isn't as big as it could be.

Artefact Problem Statement

Going back to the introduction there were a few things that the artefact was supposed to address such as the hardware advantages/disadvantages, does it change the players immersion and experience, the performance of the controller was it sustainable and ergonomically usable. The controller design was good the 8 out of 13 total testers agreed that the design of the 3D printed case could be improved such as making the design smaller so it's more user friendly, improvements to the button could be changed to be much larger and more stable since the 6x6 button that was being used wasn't properly connected to the breadboard any hard presses could push the button out because of no surface so the players were pressing on the actual button, also fixing the delay and drift when moving too far, the gyro would get misaligned with the cursor on screen making it difficult to track and defeat the enemies within the game environment.

6.2 Validation and Testing

Multiple tests were performed keeping the data anonymous the testers were sat down and read an information sheet about the project as well as signing a consent form to inform that they agree with being a tester. Once the documents

were read and signed, they were given the custom based controller and the game to play after testing they would also be required to take part in an online form which would have a variety of questions as well as opportunities for them to give thoughts about the artefact.

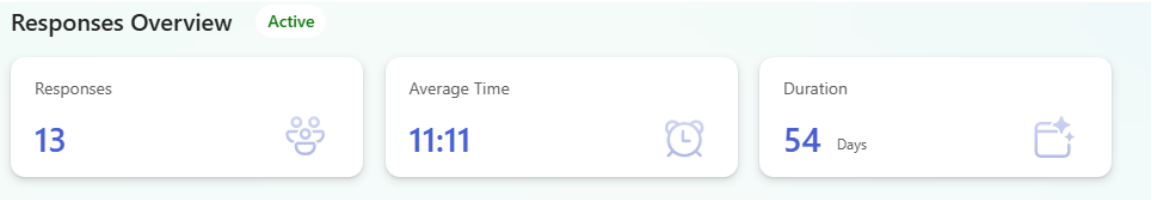

Figure 13. 13 Total Testers

There were questions based on the controller as well as the game where the controller was tested in, but the main results were from the questions surrounding the motion-based controller and how those results could be helpful to understand alternative game controllers. One of the questions were “would you use this motion-based controller instead of a traditional one?” and the results were split but most the testers went with depends.

Figure 14. Would they use the controller Question

A lot of the answers were that it would depend on the game they would play and if it makes sense, this could be related back to an earlier in the immersion section of the report where it would be beneficial if the game that players played would have supported for the controller and give them an advantage or make it fun to use. So alternative game controllers are usable, and players do like to use them, but they would need a motive to use them.

The next set of questions are mainly about the performance of the controller. “How do the controls feel overall?” 46% said that the motion-based controller was okay because of the delay and drift that the gyro had when moving to far from the screen and this would mean the controller wouldn’t be correctly moving the cursor on screen and making it awkward. The 5 testers 38% said that the controller was smooth for the most part there wasn’t any unnecessary jitter and the controller responded to fast movement and was accurate when moving.

Figure 15. Controller Feel Question

The next question was “how responsive are the controls? 1 being not at all and 10 being very responsive” And the average rating of this was higher than expected with a 7.58 Rating out of 10 which shows the controller for the most part is very responsive and with further upgrades could be improved. All 13 user testers in the small sample reported on the form that the controls were very easy to learn.

A lot of the testers also thought that the controls were good and well-made but common answers like the gyro seem to be misaligned when moving too much, and once they learnt the recentre mechanic that was implemented to help with the gyro delay and drift it made the experience better. But fixing these issues could potentially improve the experience even more.

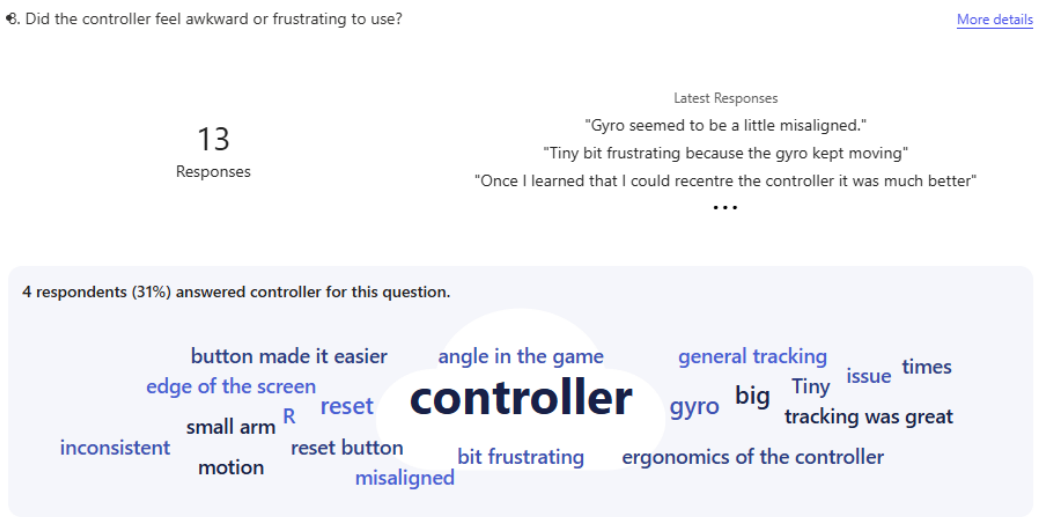

Figure 16. Controller Responses

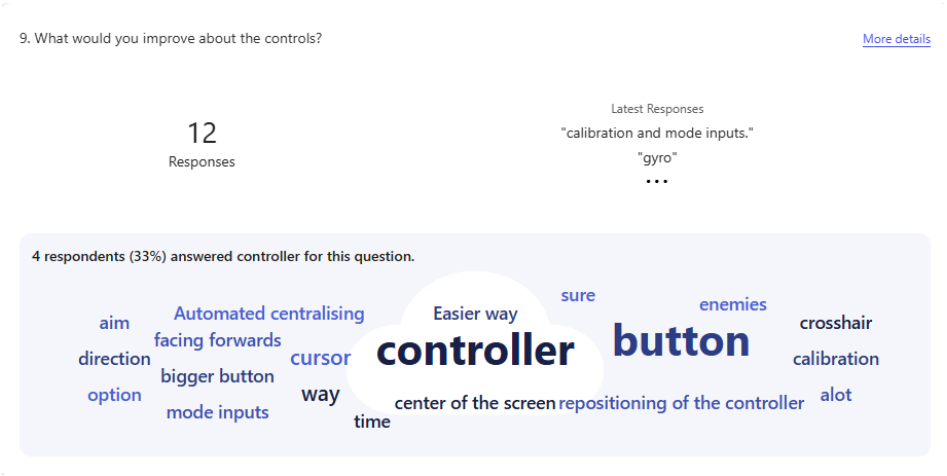

Figure 17. Controller Improvements

Figure 18. Design of the controller

10 out of 13 77% of testers said that they were somewhat confident using the motion-based controller and the other 3 testers 23% said they were extremely confident indicating people are aware of this technology and with further improvements the controller can be really fun to use further implying that companies should invest into alternative game controllers as there is a massive group of players in the market that enjoy and it is still on the arise. The average rating overall from all testers came to about 7 /10 being the performance of the controller with the game was very good and the environment to test the controller was excellent.

The next question is very important as it allowed to answer one of the main questions in this research and that's if alternative game controllers can help

shape player experience and immersion and 69% of the testers said that it did feel immersive.

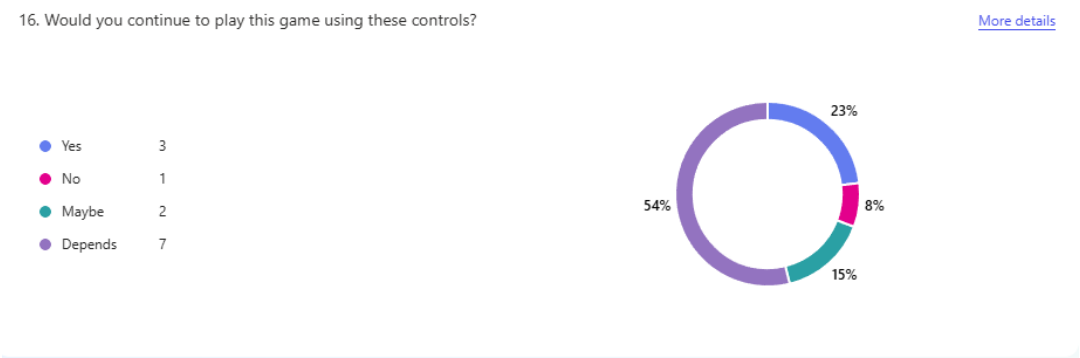

Figure 19. Immersion Question

A lot of the testers said that they would consider exploring alternative game controllers. And that overall, the experience using a custom motion-based controller was 3.82 out of 5, showing that the controller did improve the game for them and that it was very impressive and once the issues were fixed it has massive potential to be extremely fun and useful in a lot of games.

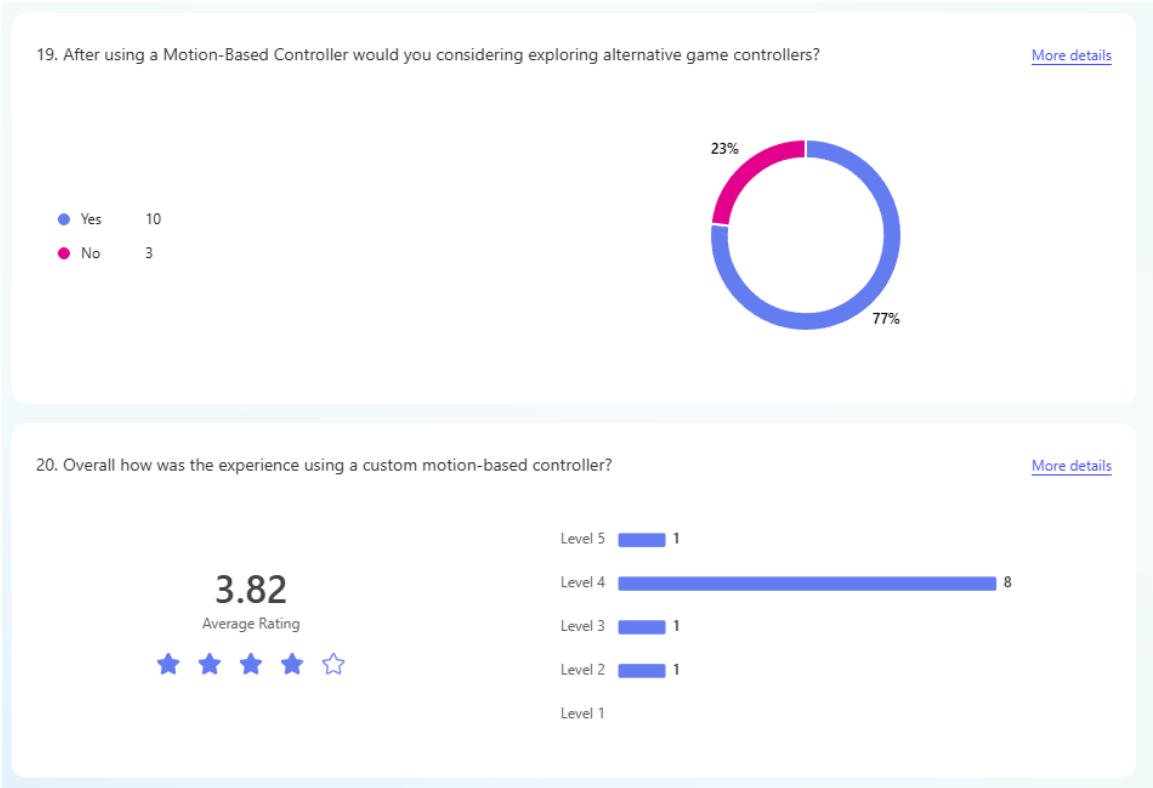

Figure 20. Experience and Exploring Questions

The user participants also answered the question on if they have every experienced any other games with similar controls and how the custom motion-based controller felt compared to those other alternative controllers. A lot of the users said that the control felt like an old school Wii game or arcade shooter, and that the controls felt smooth and because of prior experience with this technology it was pretty easy for them to use but it also came with the fact that it was less accessible and not as much supporting options.

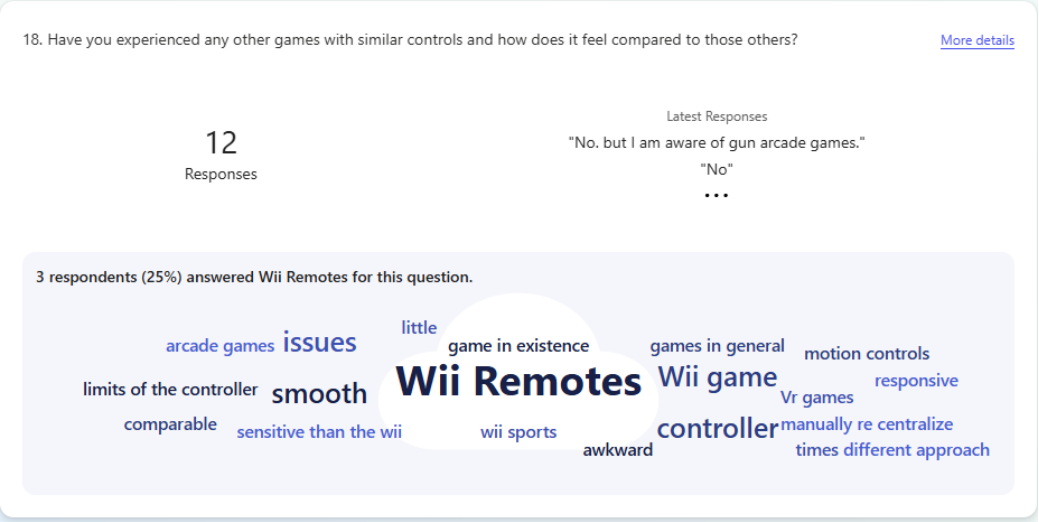

Figure 21. Comparison to other technologies similar

One of the questions that the 13 total users were asked to answer was if the controller felt meaningful or unnecessary 1 being meaningful and 10 being unnecessary and the average rating was level 5 as 5 out of the 13 testers said thought it was both unnecessary but also meaningful, this goes back to earlier when talking about how users felt that the alternative game controllers were enjoyable when the game being played made sense for them to use motion-based controls.

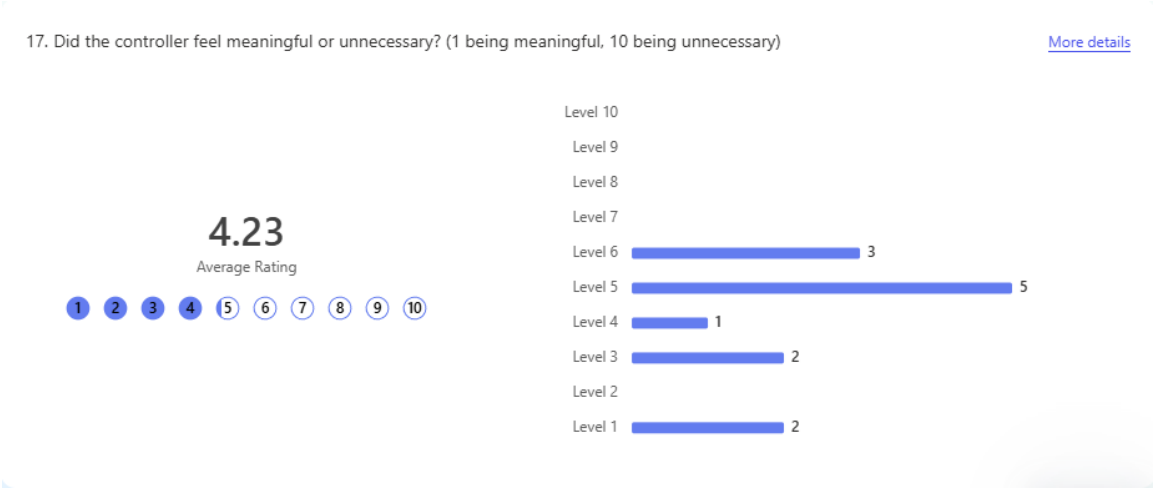

Figure 22. Was the Controller meaningful or unnecessary

Lastly the final question was based on what the testers thought they could improve or change after using the controller in game. The 4 testers 44% said that the button could be improved to become more stable as well as bigger to feel more like a controller as well as reduce the size of the 3D printed case and of course improving the gyroscope to become more accurate and less drift.

Overall, the tests confirm that the motion-based controller is both functional and fun to use answering the questions that was researched about alternative game controllers and the artefact itself was designed well to work with the game environment that was created. While there were some limitations and improvements like the gyroscope and controller design, they did not prevent the testers from using it effectively. Instead, they show further future improvements and the advantages of motion-based controllers and alternative controllers.

7.0 Critical Evaluation

The artefact is a handheld motion-based controller that allows the user to control the cursor on screen from any laptop or desktop using physical movement. It uses an MPU6050 gyroscope & accelerometer and a Raspberry Pi Pico WH to detect and process the motion with a button for additional input.

Success of the artefact

The artefact was successfully completed within the timeframe and was able to achieve its core objectives. Testing showed it was functional, responsive and

easy to learn with most testers agreeing that it improved player experience and immersion and they were able to use it effectively.

Significance

The artefact is significant as it helps understand a new way to play with alternative game controllers, it effectively demonstrates that motion-based controllers can enhance and engage player experience and immersion. In addition, it can also show that these systems can be developed using low-cost, accessible components to help stop damage against the environment and help players who may want to use this technology to help them game if they are not able to or want to use traditional controllers.

Comparison to similar technology

The artefact compared to commercial technology shows that its more affordable and flexible to build and use, working as a general HID system. However, it lacks the precision and advanced calibration that are found in VR systems and other alternative game controllers because of less components and cheaper options resulting in lower accuracy and drift issues.

Advantages and contributions

Now that the project is complete and with tests conducted, the motion-based controller has several advantages. Firstly, because of the cheap components that was purchased, the project itself was low cost, so the artefact itself didn't have much wasted materials and had all the correct components to function. In addition, it was also very simple to setup and understand once all the components arrived, making it accessible to a wide range of people compared to commercial alternatives that are much more expensive reducing the financial barrier that is often associated with gaming and especially hardware that is required to feel more immersed and bring new experiences. Furthermore, the artefact can be carried around without any issues, and the design can be easily

changed later giving it more flexibility.

New Ways to Play: A Study in Alternative Game Controllers

Main Items	Cost	Link
BreadBoard	£8	https://www.amazon.co.uk/dp/B0DB7GX4KB?psc=1&ref=ppx_pop_dt_b_product_details
JumperWires	£6	https://www.amazon.co.uk/dp/B07FBCJ7GK?ref=ppx_pop_dt_b_product_details&th=1
6x6 Buttons	£9	https://www.amazon.co.uk/dp/B0CJY2PV1J?psc=1&ref=ppx_pop_dt_b_product_details
3D printed case	Free	Printed For free
Pi Pico WH	£6.70	https://thepihut.com/products/raspberry-pi-pico-w?variant=41952994787523
MPU6050	£8.59	https://www.amazon.co.uk/dp/B07XRK5FHP?psc=1&ref=ppx_pop_dt_b_product_details
Soldering Pen	£25 + £10 shipping	https://www.fnirsi.com/products/hs-02?variant=45192934850778&country=GB&currency=GBP&utm_
	Total Cost = £73.29	

Old components for prototype	Cost	Link
Jumper Wires	£6.99	https://www.amazon.co.uk/dp/B01EV70C78?ref=ppx_pop_dt_b_product_details&th=1
6x6 Buttons	3.99	https://www.amazon.co.uk/dp/B00TXNRHT2?psc=1&ref=ppx_pop_dt_b_product_details
	Total Cost = £11	

Final Cost of Project = £84.29

Figure 23. Cost of Artefact

The artefact also offers a new way to play offering players who want to experience a different form of gaming, this also leads onto the fact that not everyone is able to play games so making a controller that supports more overall movement like moving the whole arm to look around can be immersive and beneficial. The main contribution was to create an easily accessible alternative game controller for everyone showcasing that you don't need to spend a lot of money and gives players who want and need alternative game controllers but also keeping the same functionality to be usable as well as contribute to ongoing research on alternative game controllers and with the tests and this whole project is rich information to the industry.

7.1 Impact

The artefact has potential impact on gaming in a whole massively changing how the player can play games clearly demonstrates the uses of alternative game controllers. It addresses the need for more immersive and engaging experience, especially for people who want to use something else but the traditional controller or keyboard and mice setup.

For many people seeking alternative game controllers means having a more interactive and physical way of playing video games. The test clearly shows that users are open to using alternative game controllers if it feeds their needs as well

as helping people who may not be able to use traditional controllers or inputs because of limitations or disadvantages so having a controller that doesn't require much where they can still feel immersed and have fun like anyone else is important.

From the industry view, the artefact shows the constant trends towards alternative game controllers especially motion control and immersive technology. Demonstrating that the artefact can be developed almost the same way but just much more cost effective and more accessible.

The artefact brings a new perspective to academic discussions as it contributes to usability and motion tracking technology and providing practical evidence and clear study and research into alternative game controllers. And identifying the clear issues with using cheap components such as drift and calibration challenges.

8.0 Conclusion

The project aimed to investigate whether a new way to play with alternative game specifically motion-based controllers could improved immersion, player experience and accessibility. Participants reported a positive immersive experience with the motion-based controller from the small group of testers that took part in the project testing. The problem identified was that most standard controllers do not always support for all players and don't meet expectations and limit immersion and accessibility for some users. This artefact was created to give players a new experience as well as a controller for anyone to use.

To address this, low-cost components were purchased to develop the motion-based controller and specifically designed in a way that it could be supported and tested in a Unity based game for the controller to showcase the capabilities and functionality. The project followed a structured approach with in-depth planning, research, design and development for both hardware and software and with clear user testing.

Overall, the project can be considered successful as it met its core objectives and demonstrated that motion-based controllers and alternative game controllers alike can provide an engaging and immersive experience to traditional controllers. As well as giving great data on players thoughts on alternative game controllers for future research and development.

8.1 Limitations

This project has several limitations that has affected the validity and generalisability of the results. Firstly, underestimating the scope of the project and not properly educating on the subject on the matter when it comes to the components. When designing the artefact it was hard to figure out how the final design of the controller would end up being initially starting with the idea that it would be a wearable glove with the components attached and sensor triggers on the fingers to aim and shoot but that seemed way too far for the skills that were at hand. So having the switch to a simpler idea that was more possible to achieve within the time given to us.

Secondly the sample size used for testing was relatively small, which limits how representative the findings are of a wider player population. This means that user feedback may not fully reflect broader user experiences. The project also didn't have any direct comparison with traditional controllers like a standard gamepad making it difficult to clearly evaluate whether the developed artefact provides a measurable improvement in suability or immersion. As a result, conclusions are based on linear tests rather than comparative analysis.

In addition, technical constraints were encountered, particularly relating to input responsiveness and latency. So having to find a way to mitigate these issues were a challenge but being able to add small solution by giving the game a key bind where the screen would reset back to centre was a good way to fix the gyro drift. However, they still have affected the overall user experience and performance during testing.

8.2 Future Work

Future work will focus on improving both the technical performance and evaluation of the controller. Firstly, the current MPU6050 sensor could be replaced with a sensor-fusion IMU such as the BNO055. This would reduce the drift and improve motion accuracy, addressing the responsiveness and reliability issues identified during testing.

Secondly conducting tests where there could be comparisons to study between the artefact motion-based controller and traditional controller, such as an Xbox gamepad. This would allow for a direction measurement of differences in usability, immersion, and performance, addressing the lack of evaluation in this project. To strengthen the accessibility focus, future testing should include participants with motor impairments or varying physical abilities. This would provide more meaningful insight into how effective the controller, rather than relying on assumptions based on able-bodied users.

Further improvements could include the addition of haptic feedback like rumble motors to enhance user experience and provide more immersive interaction. Additionally, replacing the current wired USB connection with Bluetooth connectivity would improve usability, portability an overall user convenience.

Finally, if there was more time now with the experience of creating an alternative controller future version of the motion controller could expand testing across multiple games and applications and focus on more complex game environments to better evaluate the controller adaptability. In addition, improving the design of the controller and moving to a more stable form and more aesthetically pleasing to use with much more improved electronic components while still being low cost and affordable.

Appendices

A. Project Timeline

Figure 24. Gantt Chart Project Timeline

B. Feasibility, Risks, and Ethical Aspects

B.1. Revised Feasibility

Having chosen this project for alternative game controllers meant that the existing knowledge for code and using other types of controllers meant the project would be much easier to complete and research in-depth. Having known about the use of Unity and Python from previous years this meant a solid starting foundation to being with to produce a motion-based controller that would fit the project plan. Creating Gantt charts and the use of Kanban on GitHub this meant

a solid start in the project with good research and information before moving on to the development phase of the artefact and complete the project with the best results.

B.2. Updated Risk Analysis and Mitigation

Potential Risk	**Likelihood (Low/Med/High)**	**Impact (Low/Med/High)**	**Mitigation Strategy**
Hardware malfunctioning, due to faulty electronic components or damaged	*Medium*	*High*	*Identify the best Components and researching previous user experiences to ensure best possible outcome.*
Errors in the code when developing a seamless interaction between motion controller and Unity	*Medium*	*Medium*	*Allocate time into tutorials and guides and seeking guidance with supervisor.*
Finding participants that are willing to take part in the project as test	*Low*	*High*	*Look for individuals that are willing to take part in testing the product, online or real life early.*
Possible Participant withdrawal during testing	*High*	*High*	*Get as many testers as possible in case a few do withdraw there is still enough*
Component Delivery delays when purchasing online	*High*	*High*	*Early planning and research to purchase everything on time.*
Issues with GitHub possible loss of data and Version Control conflicts	*Low*	*High*	*Ensure all the work is backed up and saved and pushed into GitHub at the end of the session.*
Possible illness	*High*	*High*	*Ensure to stay fit and healthy during*

during the project could delay how much time spent working on artefact			*the project and not spend too much time out or overworking.*
Ordering the incorrect components	*Low*	*Medium*	*Not likely to happen but if it did, getting a refund is always the option.*
Components malfunction because they weren't implemented correctly	*High*	*High*	*Ensure all the components are properly soldered and the jumper wires, MPU6050 and Pi Pico are pushed all the way down.*

Table 4. Risk Assessment and Mitigation

B.3 Revised Ethical Considerations

During this project, there will be a lot of aspects of human interaction and having to make sure that the development of the project needs to be ethically considered, as well as how this can also be thought through. This project will ensure that the products being used are ethically sustainable, such as electronic components and Unity software that are free and at a low cost. The data handling will be done online as this is a much easier process. The project will follow academic guidelines and any information used will be correctly credited.

Furthermore, the data that was collected during the user testing will ensure its all consenting and will all be anonymous for the online forms as they weren't asked to use emails or provide names and the consent forms that were signed before and after reading the information sheet will be kept private and can be accessed by them if they would like to withdraw.

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